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AI Medical Diagnosis Application

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Abstract-----Artificial Intelligence aims to mimic the human cognitive powers. AI is also into healthcare where it's been used for automating the diagnosis processes powered by increased availability of healthcare data. This research work develops an application, which can be used in diagnostic centers for multiple diseases diagnosis like Arrhythmia classification using ECG Image analysis, Malaria using Cell images, Brain tumor detection using MRI Scans and many more. At Present, this research work attempts to develop an Arrhythmia classification using ECG Image analysis and would add other diagnosis later on. Basically, ECG signals are used for analysis of Arrhythmia. The irregularities of heart beat are commonly called Arrhythmia. Here, a Convolutional Neural Network model is proposed for the classification of signals from ECG images to different classes. There are many different classes which are normal beat, Premature ventricular contraction beat, paced beat, atrial bundle branch block beat, left bundle branch block beat, atrial premature contraction beat, ventricular flutter wave beat, and ventricular escape beat, etc.

Keywords ---- Diagnosis, Convolutional Neural Networks, Arrhythmia, ECG signals, Classification, Deep Learning, Web Application.

I. INTRODUCTION

Using AI, the healthcare people or professionals can analyze the patient or people health data precisely for early diagnosis of disease and better treatment. Today, it's possible to say to say if a person has a skin cancer or not from just a selfie or picture using machine learning and computer vision. Similarly, heart arrhythmia classification, brain tumor detection, lung cancer detection, Diabetes Prediction and a lot more can be done in minutes with very good performance and more precisely than humans. In diagnostics, radiologists and many more human powers is needed to analysis the ECG signals, MRI scan etc. It really requires a lot of human power which can be substituted with very little less errors using the automated AI application, this research work can predict or diagnose the disease within seconds. The proposed research attempts to get more than 99% accurate results in the predictions within seconds leading to early checkup and treatment. At Present, there is arrhythmia classification in the proposed application.

The leading cause of human death is Cardiovascular diseases (CVD), with millions of people lose their life annually due to CVD's [1]. CVD's classification could reduce the mortality rate providing the timely medicine. In arrhythmia, heartbeat patterns deviate from the regular/normal heartbeat. Arrhythmia does not necessarily mean slow or fast heartbeat, it means that the heart beat is having irregular heart beat patterns. Our model basically classifies the different arrythmia and will help in diagnosing and treatment of heart diseases. Detection of such patterns automatically is of great use in medical practice [1].

Recording ECG signals manually is a very time-consuming task and it needs more effort. Detecting these arrhythmias automatically and classification assist physicians very much [2,3]. Detecting these cardio vascular diseases early can be helpful for proper treatment. Many techniques and procedures were proposed for detection of arrhythmia [4-10].

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For detecting Arrhythmia, Clinical experts might check the ECG signals for the longer period of time.

We take ECG images, apply 2D transformations on the images. That transformed dimensional spectrogram is obtained by converting 1-D signal using short-time Fourier transform (STFT) and combine both the frequency and time information into a single matrix. This matrix of spectrogram coefficients is useful for getting useful features for ECG signal representation. This matrix representation allows the application of many different CNN architectures useful for developing automated systems.

We have our application which has a lot of diagnosis options where we could choose one for prediction or detection. After we choose a disease to detect, we click on browse and upload an image. On clicking analyse Image, within seconds it renders the output.

II. LITERATURE SURVEY

Accurately detecting and analysing the clinical disease through ECG, MRI scan, CT Scan etc is a challenging task. It requires doctors, radiologists to analyse them and detect any disease which often takes a lot of time. Previously there are few applications which has few diseases which can be diagnosed.

A. *Saliency AI*

Saliency AI is an application which is used to leverage deep learning techniques in research and development. This Saliency automates the medical workflow. It halves our time for preparation of images for physician review and image annotations. This application has Lesion's validation, Brain MRI, Knee Osteoarthritis, Deidentification. This automated application reduces the human errors with great accuracy.

B. *Ezra*

Ezra is an application which is used for early cancer detection. This is an application which leverages AI detecting the cancer so that people can get treated earlier. This automated application reduces human errors and in fact detect if there is cancer in any part of our body quickly and accurately.

C. *SkinVision*

This application used for skin cancer detection. In this application we can find if a person has skin cancer or not just by taking photos or selfies of our skin. This automates the process of diagnosis of skin cancer.

D. *Arrhythmia Classification*

Many literatures were proposed for the arrhythmia classification and every proposed literature has its limitations. The different methods proposed in the literature are frequency analysis [17], heuristic-based methods [18], ANN [19], SVM [20], statistical methods [21], wavelet transform [22] and mixture of expert systems [23]

This [11] paper proposed a method that classifies 16 different arrhythmias. Features are extracted using these two namely ICA and wavelet transform and then ECG features were obtained by calculating RR interval. Then these features were used for classification of arrhythmia using Support Vector Machines (SVM) with accuracy of 99.6%.

Got an accuracy of 96.95% which used Feed Forward Network as classifier and PCA and wave transform as extracting features and used c-means clustering for reduction of features [12].

Using SVM with particle swarm optimization, average of 89.72% accuracy is obtained [13].

Got an accuracy of 98.06% using Recurrent Neural Network (RNN) which used eigen vectors for feature extraction [14].

Using 1-D CNN for classification of features from 1-D CNN 96.73% accuracy is obtained. Deeper 1-D CNN is used and 97.04% accuracy is obtained [15]. In these cases, large dataset was used. A 2-D CNN model with nine layers was used and got accuracy of 94.03% [16].

Although lot of literatures were proposed, there are many limitations. 1) Lower classification performance in practical. 2) Good performance on carefully selected ECG recordings only. 3) Less or limited number of arrhythmia types for classification. 4) classification accuracy on less occurring arrhythmia classes was not good. 5) Performance of model is tested only on lower datasets and not on larger datasets.

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III. PROPOSED SYSTEM

Here we are implementing a complete application which has a lot of diseases to be diagnosed. Entire application follows the following architecture.

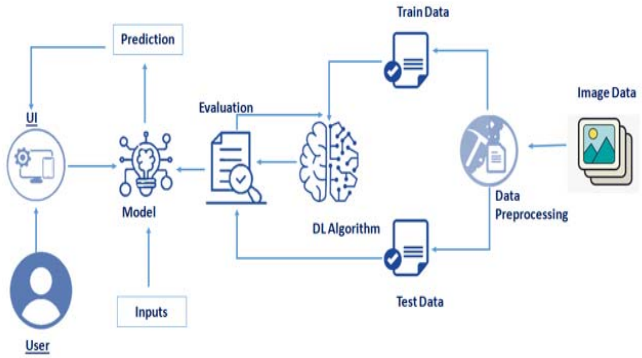

Fig.1 overall Architecture

This architecture is for a single disease and similarly we are adding multiple diseases implementation deployed in a single application. We take an Image data or a dataset, we perform preprocessing upon the dataset which includes removing the noise from the dataset, cleaning of the data and making the data into a form which is used for modeling. We divide the data into train and test data. Almost 70% or 80% of the data is divided into training data for training the model and 30% or 20% of the data is testing data used for testing. We use Train data and suitable algorithm for training the model and evaluate the model using Test data. We use this model for future predictions and that predicted output is seen on the application. Our Arrhythmia Classification method consists of preprocessing, generation of 2D images, Data augmentation feature extraction and finally classification.

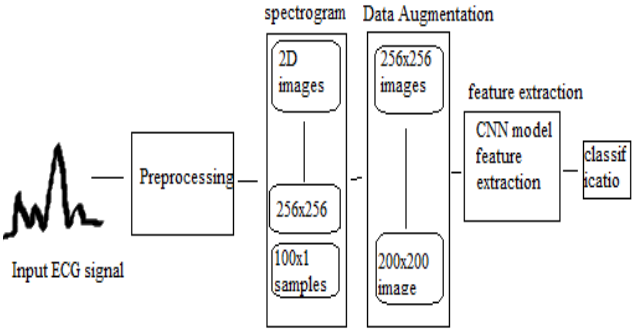

Fig.3 Complete procedure of ECG signal classification.

A. Data Collection

We collected our data from MIT-BIH arrhythmia classification which has 48 records each of duration 30 minutes collected between 1975 and 1979. The data is publicly available in MIT-BIH website

B. Data Preprocessing

Basically, there is noise in the ECG signal, our moto here is to remove the noise. We need to remove the 3 types of noise from the ECG signal to make sure that denoised ECG signals are used for model building and processing. We did combine reconstruction algorithm of wavelet decomposition and wavelet-based thresholding to remove noise from ECG signal. ECG images are obtained by transforming ECG signals in preprocessing step.

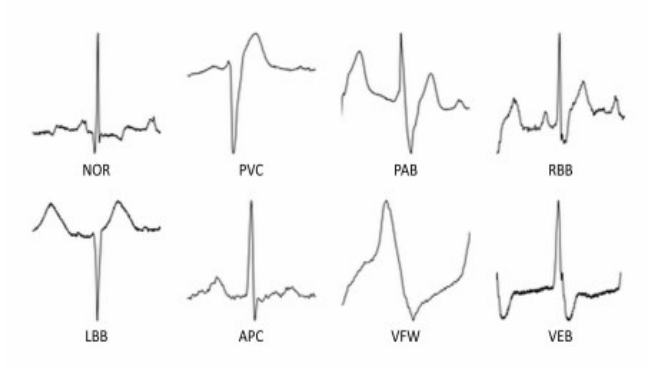

Fig. 2 Normal beat and seven ECG arrhythmia beats

C. Generation of 2D images

For time series analysis, 1-D CNN can be used but it is limited to 1-D kernels or filters. We can also use 3-D CNN but it takes lot of time and need huge amounts of data and even resources. We can use 2-D filters/kernels compared with 1D kernels and 3D kernels. By using 2-D CNN, we can convert 1D signals to 2D signals. By using short-time Fourier transform (STFT), we can generate 2D time frequency spectrograms.

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D. Data Augmentation

For 2-D ECG signals, data augmentation produces more amount of data and learn more variations and is very useful. Data augmentation is used for generating more amount of training data. The ECG signal is highly imbalanced, there is more data with normal heartbeat so as to balance the dataset, we oversample the imbalanced data using data augmentation.
When we have an imbalanced data, we can use data augmentation for having balance between classes. Another problem is overfitting. When there is small dataset, it leads to overfitting. In order to overcome this, we can overcome this by using data augmentation and hence allows better training of CNN.

E. Deep CNN

The below is the architecture of our proposed convolutional neural network (CNN)

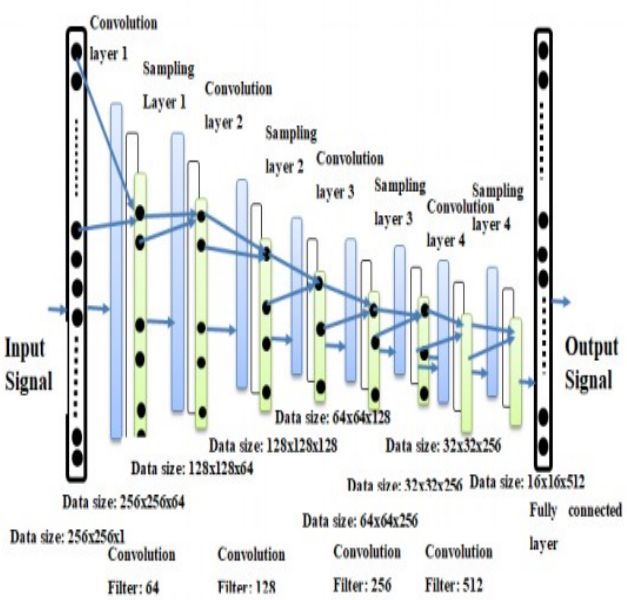

Fig.4 Architecture of proposed CNN

The model has a CNN architecture with five 2-D convolutional layers and we have a pooling layer next to each convolutional layer. Since our model is a multiclass classification, we are using SoftMax activation in the output layer.
At last, fully connected layer is used before output layer.
The performance of CNN architecture which is been described in proposed system, is compared with AlexNet and VGGNet.
The details of architecture of proposed CNN is mentioned in Table 1, architecture of Alexnet is mentioned in Table2 and architecture of VGGNet is mentioned in Table3.

Layers	Type	Filter size	Stride	Kernel	Input size
Layer1	Conv2D	3x3	1	64	256x256x1
Layer 2	Pooling	2x2	2	-	256x256x64
Layer 3	Conv2D	3x3	1	128	128x128x64
Layer 4	Pooling	2x2	2	-	128x128x128
Layer 5	Conv2D	3x3	1	256	64x64x128
Layer 6	Pooling	2x2	2	-	64x64x256
Layer 7	Conv2D	3x3	1	512	32x32x256
Layer 8	Pooling	2x2	2	-	32x32x512
Layer 9	FC	-	-	4096	16x16x512
Layer 10	output	-	-	8	4096

Table 1: Details of our proposed CNN model architecture

Layers	Type	Filter size	Stride	Kernel	Input Size
Layer1	Conv2D	11x11	1	3	128x128x1
Layer2	Conv2D	5x5	1	48	128x128x48
Layer3	Pool	3x3	2	-	128x128x48
Layer4	Conv2D	3x3	1	128	63x63x48
Layer5	Pool	3x3	2	-	31x31x128
Layer6	Conv2D	3x3	1	192	31x31x128
Layer7	Conv2D	3x3	1	192	31x31x192
Layer8	Conv2D	3x3	1	128	31x31x192
Layer9	Pool	3x3	2	-	31x31x128
Layer10	Full	-	-	2048	15x15x128
Layer11	Full	-	-	2048	2048
Layer12	Out	-	-	8	2048

Table 2: Details of AlexNet

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Layers	Type	Filter size	Stride	Kernel	Input size
Layer1	Conv2D	3x3	1	64	128x128x1
layer2	Conv2D	3x3	1	64	128x128x64
layer3	Pool	2x2	2	-	128x128x64
Layer4	Conv2D	3x3	1	128	64x64x64
Layer5	Conv2D	3x3	1	128	64x64x128
Layer6	Pool	2x2	2	-	64x64x128
Layer7	Conv2D	3x3	1	256	32x32x128
Layer8	Conv2D	3x3	1	256	32x32x256
Layer9	Conv2D	3x3	1	256	32x32x256
Layer10	Pool	2x2	2	-	32x32x256
Layer11	Conv2D	3x3	1	512	16x16x256
Layer12	Conv2D	3x3	1	512	16x16x512
Layer13	Conv2D	3x3	1	512	16x16x512
Layer14	Pool	2x2	2	-	8x8x512
Layer15	FC	-	-	2048	8x8x512
Layer 16	FC	-	-	2048	2048
Layer 17	Out	-	-	8	2048

Table 3: Details of VGGNet

IV. EXPERIMENT AND RESULTS

The proposed system CNN architecture is compared with AlexNet and VGGNet. Here we are using MIT-BIH arrhythmia dataset for evaluation of our models.

A. Experiment Setup

The proposed CNN model is implemented in python, TensorFlow and Flask. The setup consists of Acer Predator laptop with 16GB RAM, 500GB SSD, and NVIDIA 2080 GPU. The 2-D spectrogram images are divided into train and test data. Train data is of 70%, Test data is of 30%. We used 5-fold cross validation for training our model.

B. Cost Function

There are many cost functions which can be used. In our Experiment, we used cross entropy loss function. Cost function is a measure of error made by model comparing actual value to the predicted value. We need to reduce the loss. We use optimizers to reduce or minimize the error function. We used cross entropy in our model. Cross entropy loss function is given as:

$$C = \frac{-1}{n}\sum_{c=1}^{N}([y_c * ln(a_c) + (1 - y_c)ln(1 - a_c)])$$

C. Evaluation Metrics

The Following Metrics are used:

1. *Accuracy:*

It refers to the percentage of records or data points which are correct.

$$\text{Accuracy (A)} = \frac{TP + TN}{TP + TN + FP + FN}$$

2. *Precision and Recall*

Precision refers to the number of datapoints that are correctly classified as positive in the total no of points that are identified as positive.

Recall refers to number of datapoints that are correctly classified as positive in the total no of positive points.

$$\text{Precision (P)} = \frac{TP}{TP + FP}$$

$$\text{Recall (R)} = \frac{TP}{TP + FN}$$

3. *Specificity*

It refers to the True Negative Rate

$$\text{Specificity (S)} = \frac{TN}{TN + FP}$$

4. *F1 Score*

It is derived from Precision and Recall

$$\text{F1 Score} = \frac{2 \times P \times R}{P + R}$$

Here TP = True Positive
TN = True Negative
FP = False Positive
FN = False Negative

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D. *Results*

We compared results with varying batch size and learning rate. When batch size is 1000, then the accuracy of the model showed large fluctuations. When the batch size is up to 2000, the accuracy increased but it is not all stable we can say it is unstable. Batch size of 2800 reached a stable state. The table shows the results we obtained taking batch size as 2800 and changing learning rates:

Learning rate	Batch size	Accuracy
0.001	2800	99.11
0.005	2800	98.83
0.100	2800	98.88
0.200	2800	98.91

When the learning rate is very small that is less than 0.0005, then the rate of convergence is very slow but when the learning rate is high which is 0.001 then there is a faster convergence rate. The below table shows the accuracy with constant learning rate of 0.001 and varying batch size.

Learning rate	Batch Size	Accuracy
0.001	2800	99.11
0.001	2000	98.95
0.001	1000	99.00
0.001	500	98.93

We compared our proposed CNN model with AlexNet and VGGNet and got the following results with augmented and native methods.

Model	Native/augmented	Accuracy	Precision	Recall	Specificity	F1 Score
AlexNet	Native	98.85	98.59	97.08	99.62	0.97
AlexNet	Augmented	98.81	98.63	96.81	99.68	0.97
VGGNet	Native	98.63	97.86	96.93	99.37	0.97
VGGNet	Augmented	98.77	98.08	97.26	99.43	0.97
Proposed	Native	99.11	98.58	97.91	99.61	0.98
Proposed	Augmented	98.92	98.69	97.26	99.67	0.98

V. FUTURE WORK

At present we added only arrhythmia classification in our application, similarly we will add many more diseases with good accuracy. We are also trying to get an accuracy of this proposed model greater than 99.11% by adding batch normalization, dropouts and adding building deeper CNN.

VI. REFERENCES

[1] Amin Ullah ,Syed Muhammad Anwar , Muhammad Bilal and Raja Majid Mehmood, “Classification of Arrhythmia by Using Deep Learning with 2-D ECG Spectral Image Representation” Remote Sens. 2020, 12, 1685; doi:10.3390/rs12101685.

[2] P. de Chazal and R. B. Reilly, “A patient-adapting heartbeat classifier using ECG morphology and heartbeat interval features,” IEEE Transactions on Biomedical Engineering, vol. 53, no. 12, pp. 2535–2543, 2006.

[3] A. Mustaqeem, S. M. Anwar, M. Majid, and A. R. Khan, “Wrapper method for feature selection to classify cardiac arrhythmia,” in Proceedings of 39th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), pp. 3656–3659, IEEE, Jeju Island, South Korea, July 2017.

[4] A. Ebrahimzadeh, B. Shakiba, and A. Khazaee, “Detection of electrocardiogram signals using an efficient method,” Applied Soft Computing, vol. 22, pp. 108–117, 2014.

[5] G. de Lannoy, D. François, J. Delbeke, and M. Verleysen, “Weighted conditional random fields for supervised interpatient heartbeat classification,” IEEE Transactions on Biomedical Engineering, vol. 59, no. 1, pp. 241–247, 2012.

[6] S. Poungponsri and X.-H. Yu, “An adaptive filtering approach for electrocardiogram (ECG) signal noise reduction using neural networks,” Neurocomputing, vol. 117, pp. 206–213, 2013.

[7] E. J. d. S. Luz, D. Menotti, and W. R. Schwartz, “Evaluating the use of ECG signal in low frequencies as a biometry,” Expert Systems with Applications, vol. 41, no. 5, pp. 2309–2315, 2014.

[8] Z. Zidelmal, A. Amirou, D. Ould-Abdeslam, and J. Merckle, “ECG beat classification using a cost sensitive classifier,” Computer Methods and Programs in Biomedicine, vol. 111, no. 3, pp. 570–577, 2013.

[9] R. J. Martis, U. R. Acharya, and L. C. Min, “ECG beat classification using PCA, LDA, ICA and discrete wavelet transform,” Biomedical Signal Processing and Control, vol. 8, no. 5, pp. 437–448, 2013.

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[10] A. Mustaqeem, S. M. Anwar, and M. Majid, “Multiclass classification of cardiac arrhythmia using improved feature selection and SVM invariants,” Computational and Mathematical Methods in Medicine, vol. 2018, Article ID 7310496, 10 pages, 2018.

[11] C. Ye, B. V. Kumar, and M. T. Coimbra, “Heartbeat classification using morphological and dynamic features of ECG signals,” IEEE Transactions on Biomedical Engineering, vol. 59, no. 10, pp. 2930–2941, 2012

[12] Ceylan R, zbay Y (2007). Comparison of FCM, PCA and WT techniques for classification ECG arrhythmias using artificial neural network. Expert Systems with Applications 33(2):286-295

[13] Melgani F, Bazi Y (2008). Classification of electrocardiogram signals with support vector machines and particle swarm optimization. IEEE transactions on information technology in biomedicine 12(5):667-677

[14] beyli ED (2009). Combining recurrent neural networks with eigenvector methods for classification of ECG beats. Digital Signal Processing 19(2):320-329

[15] Rajpurkar, P.; Hannun, A.Y.; Haghpanahi, M.; Bourn, C.; Ng, A.Y. Cardiologist-level arrhythmia detection with convolutional neural networks. arXiv Prepr. 2017, arXiv:1707.01836

[16] Acharya, U.R.; Oh, S.L.; Hagiwara, Y.; Tan, J.H.; Adam, M.; Gertych, A.; San Tan, R. A deep convolutional neural network model to classify heartbeats. Comput. Biol. Med. 2017, 89, 389–396.

[17] Minami, K. I.; Nakajima, H.; Toyoshima, T. Real-time discrimination of ventricular tachyarrhythmia with Fourier-transform neural network. IEEE Trans. Biomed. Eng. 1999, 46, 179–185.

[18] Osowski, S.; Hoai, L. T.; Markiewicz, T. Support vector machine based expert system for reliable heartbeat recognition. IEEE Trans. Biomed. Eng. 2004, 51, 582–589.

[19] Coast, D.A.; Stern, R.M.; Cano, G.G.; Briller, S.A. An approach to cardiac arrhythmia analysis using hidden markov models. IEEE Trans. Biomed. Eng. 1990, 37, 826–836.

[20] Mustaqeem, A.; Anwar, S.M.; Majid, M. Multiclass classification of cardiac arrhythmia using improved feature selection and SVM invariants. Comput. Math. Methods Med. 2018

[21] Willems, J. L.; Lesaffre, E. Comparison of multigroup logistic and linear discriminant ecg and vcg classification. J. Electrocardiol. 1987, 20, 83–92.

[22] Hu, Y.H.; Tompkins, W.J.; Urrusti, J.L.; Afonso, V.X. Applications of artificial neural networks for ECG signal detection and classification. J. Electrocardiol. 1993, 26, 66–73.

[23] Hu, Y.H.; Palreddy, S.; Tompkins, W.J. A patient-adaptable ECG beat classifier using a mixture of experts approach. IEEE Trans. Biomed. Eng. 1997, 44, 891–900.

[24] G.SusmithaValli " An approach to detect Bone Tumor Using Segmentation Technique" IJPR; ISSN No:0975–2366; 2018

[25] K Sai Prasad, N Chandrasekhar Reddy, Puneeth B.N "A Framework for Diagnosing Kidney Disease in Diabetes Patients Using Classification Algorithms. SN COMPUT. SCI. 1, 101.

[26] Ranganathan, G. "Real Life Human Movement Realization in Multimodal Group Communication Using Depth Map Information and Machine Learning." Journal of Innovative Image Processing (JIIP) 2, no. 02 (2020): 93-101.

[27] Smys, S., Joy Iong Zong Chen, and Subarna Shakya. "Survey on Neural Network Architectures with Deep Learning." Journal of Soft Computing Paradigm (JSCP) 2, no. 03 (2020): 186-194

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